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Research Study

Hammill Institute on Disabilities

Remedial and Special Education
2025, Vol. 46(1) 53–65
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DOI: 10.1177/07419325241234079
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Comparison of Using Modified and Nonmodified Books on Comprehension of Students With Extensive Support Needs

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Abstract

Modified books are frequently used as comprehension supports for students with extensive support needs (ESN), despite limited evaluations of their creation process or impact on comprehension. This study evaluates the impact of individualized book and comprehension question modifications made through a systematic decision-making process on student comprehension during nonfiction shared book reading in general education classes. A repeated acquisition single-case research design was used to compare student comprehension of nonmodified and modified books for one first-grade student and one fourth-grade student with autism and moderate intellectual disability in a suburban U.S. Midwest elementary school. Findings indicate student comprehension was higher after reading modified grade-level, nonfiction books compared with similarly matched nonmodified books. These findings show students with ESN can access grade-level books in general education settings with individualized modifications. Recommendations for research and practice are shared. These recommendations include a systematic process for identification of the type and intensity of book modifications that may support comprehension for students with ESN.

Keywords

literacy, book modification, extensive support needs

Federal law requires that students with disabilities, including those with extensive support needs (ESN), such as intellectual disability, autism, and multiple disabilities and have pervasive support needs across domains, are provided access to and make progress in the general education curriculum (Individuals with Disabilities Education Improvement Act, 2004). This mandated access to the general education curriculum includes literacy curriculum and instruction. Literacy, a valued skill for all people, includes listening, speaking, interacting, reading, writing, and spelling (Downing, 2007). Literacy enhances and creates opportunities for students to interact in their home, community, and school environments, as well as to engage within established adult roles, such as employment (Ruppar et al., 2017). Acquiring literacy skills is paramount to ensuring all students can participate in the modern world that requires constant engagement with text in school, work, and the community (Keefe & Copeland, 2011).

Despite its importance, literacy instruction has not always been considered an integral component of education for students with ESN (Browder et al., 2011). Recent changes in federal legislation, such as the Every Student Succeeds Act (2015), have led to an increase in research of evidence-based literacy instruction for students with ESN. Unfortunately, most of this research was conducted and continues to occur in segregated special education settings (Toews & Kurth, 2019). This is concerning, considering students with ESN are most likely to access the general education curriculum and literacy instruction when they are included in general education classrooms (Ruppar et al., 2018). In addition, relying on studies that occur in separate settings is problematic, given students with ESN who have access to general education curriculum within general education settings experience academic, social,

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Associate Editor: Erin Barton

and vocational benefits during school and adult life (Kurth & Mastergeorge, 2012; Ryndak et al., 2010). Many researchers studying literacy for students with ESN in special education settings call for the replication of their investigations in general education settings (e.g., Browder et al., 2011; Mims et al., 2009), yet few have directly investigated literacy interventions in general education settings (Toews & Kurth, 2019). Without implementation of interventions in general education settings, conclusions about their efficacy in that environment are unknown.

Given federal law requires student access to general education curriculum, the dearth of research available on how to provide access to that curriculum for students with ESN in general education settings is problematic. This paucity of research has resulted in educators reporting that they do not know how to succeed in making academic curricula accessible to students with ESN (Matzen et al., 2010; Saloviita, 2015). In addition, there are noted challenges in the provision of physical access to general education classrooms without the necessary supports to make the curriculum cognitively accessible (Ryndak et al., 2009). For example, students may receive placement in the general education setting, but not receive the supports necessary for them to engage in the content covered. Thus, future research related to accessible literacy instruction strategies for students with ESN should be conducted in general education settings.

Book Modifications in Shared Reading

Shared reading is an instructional strategy with moderate research evidence to support vocabulary, comprehension, and emergent literacy skills for students with ESN (Toews et al., 2021; Hudson & Test, 2011). Shared reading is a book-based interaction between a reader and listener in which the reader presents the book and pauses intermittently to engage the listener in interaction with the text through discussion, question, and response, or physical interaction. A recent literature review of shared reading research for students with ESN identified the use of modified books as a useful strategy to facilitate student access to instruction or responding (Toews et al., 2021). Trends identified in effective book modifications include adding simplified text, visuals within books and question response options, and repetitive text lines (Toews et al., 2021).

Repetitive text and simplified text structures can help students focus on and identify the main concepts or story events in a book. Repeated readings of the same line give students multiple opportunities in the same book to comprehend the main idea and hear new vocabulary in different contexts (Mims et al., 2009). Repetitive text structures have been used extensively in research on shared reading interventions with students with ESN to support student vocabulary acquisition, engagement, and book comprehension (e.g., Spooner et al., 2014). Simplified text or adapted text has also been shown to support literacy skill development for students with ESN. For example, Ruppar et al. (2017) rewrote the classic novel, *The Odyssey*, using a systematic process to reduce the complexity of text. These researchers embedded shared reading of the adapted novel within typical instruction in a ninth-grade general education English class that included one student with ESN including hearing loss and complex communication support needs. A functional relation was established between the intervention and an increase in student vocabulary knowledge, engagement, and comprehension.

Although modified books have supported student comprehension within studies on shared reading, no studies have compared student comprehension of modified versus nonmodified books within shared reading. Understanding the extent to which book modifications affect student comprehension during shared reading may be critically important to understand the necessary components of literacy interventions for students with ESN. In addition, a recent review of the shared reading literature (Toews et al., 2021) noted only two studies investigated the use of shared reading using modified books for students with ESN delivered entirely in general education settings (i.e., Courtade et al., 2013; Ruppar et al., 2017), indicating a need in the field for further investigation of the intervention in general education settings.

Access to Content Instruction Through Shared Reading

Components of effective literacy interventions, such as shared reading, also need to occur in the context of grade-level curricula. Grade-level content can be delivered through book reading activities to promote the acquisition of content knowledge through nonfiction books (Common Core State Standards Initiative, 2010). Nonfiction books are frequently used across content area instruction (e.g., science, social studies, journalism); thus it is critical that students receive instruction in comprehension strategies using nonfiction materials (Knight et al., 2018). Little is known about effective grade-level nonfiction book modifications for students with ESN in general education settings, despite the utility of these modifications as a vehicle for learning. Nonfiction books can provide information for students through text features (e.g., captions or diagrams) and written text. As students transition to reading to learn and acquire novel information, providing content-specific books that include textual modifications (e.g., repeating key vocabulary; Ruppar et al., 2017) may become a critical support for students with ESN to make progress in the general curriculum. However, the limited evaluations of book modifications prohibit educators' understanding of how to create effective modifications to promote student comprehension of content information in nonfiction books.

The dearth of research on effective practices such as book modifications limits educator knowledge of effective instructional practices for students with ESN and thus creates a barrier to inclusion of students with ESN in general education settings (Matzen et al., 2010; Saloviita, 2015). As investigations designed to identify effective literacy interventions such as modified shared books are explored, it is also imperative to investigate the feasibility of these interventions for educators. In the case of shared reading that utilizes modified books, educators must allocate time to plan instruction and develop modified books. In addition, as research on shared reading often occurs in a one-on-one student to interventionist ratio, educators must identify time during the school day to deliver the intervention. These time considerations are important because extant research indicates book modification can be a time-consuming process (Browder et al., 2009; Shurr & Taber-Doughty, 2017) and educators may struggle to identify time for one-on-one interventions (Kim et al., 2018; Roberts et al., 2019).

Purpose and Research Questions

The purpose of this study was to provide the first investigation of the impact of modified nonfiction books compared with nonmodified nonfiction books as a support to increase the comprehension skills of students with ESN when embedded in a shared book reading intervention in general education settings. In addition, the study was also designed to gather general and special educator feedback about the feasibility and meaningfulness of modified nonfiction book shared reading in general education settings. The following research questions were investigated:

> **Research Question 1:** Do students with ESN answer more modified comprehension questions correctly during shared reading with grade-level nonfiction books with text modifications compared with nonmodified comprehension questions and nonmodified books?
> **Research Question 2:** Do general and special educators think shared reading using modified grade-level nonfiction books is feasible to implement and meaningful to their students with ESN?

Method

Participants

Following university institutional review board (IRB) human subject-, district-, and school-level project approval, educators at one elementary school in a suburban Midwestern school district in the United States were asked to identify students who met the following inclusion criteria: (a) has an intellectual disability, autism, or multiple disabilities and a concomitant significant cognitive disability; (b) attends a general education class during literacy rotations at least 3 times per week (or will attend a general education class during center rotations for the duration of the study); (c) does not miss more than an average of one school day per month; and (d) has a consistent means of responding to questions (orally, written, or through text or image selection). Consent was received from legal guardians of three students. After receiving parent consent, special and general educators participated in an in-person interview, completed a student demographic survey, and completed *The Academic & Communication Assessment Survey* (ACAS; Nash et al., 2016), an adapted version of the First Contact Survey from Dynamic Learning Maps. Interviews and surveys provided information about student support needs related to literacy, communication, and behavior. These data are integrated into the student descriptions below.

The student's special and general education teachers also participated in an interview about each student participant which included questions about (a) student present levels in literacy skills (i.e., engagement with books, phonemic awareness, phonics, fluency, vocabulary, comprehension), (b) current student literacy interventions or programs, (c) student preferred response modes (e.g., verbal, aided augmentative and alternative communication [AAC], unaided AAC), (d) student behavior support systems, and (e) any other information the teacher thought was important for a person to know who will be working on reading with the student participant (e.g., student interest or triggers for frustration). After gathering these data, one student was withdrawn from study participation because this student did not have a consistent means to communicate with others.

Ben was a 9-year-old, fourth-grade, White, male student with autism and moderate intellectual disability. Ben spoke only English at home and school. He received special education services within a separate special education classroom for most of the school day and attended a fourth-grade general education classroom during academic instructional time for approximately 1 hr a day during silent reading and science or social studies. He did not receive paraprofessional support in the general education setting and did not have a behavior support plan in his individualized educational program (IEP). Ben qualified to take his state alternate assessment. He communicated verbally in one- or two-word phrases and had an approximately 51- to 200-word spontaneous vocabulary. Ben received literacy instruction using an alternative special education curriculum in the special education class. He independently read at a middle first-grade level and had a literacy goal within his IEP to answer "wh" questions about text read to him. The study occurred during silent reading time in the general education class. Prior to the study, Ben listened to books independently on a computer program during this time.

Camilla was a first-grade, White, female student with autism and moderate intellectual disability. Camilla spoke only English at home and school. She received special education services within a separate special education classroom for most of the school day and attended a first-grade general education class during academic instructional times for 15 min a day for a partner reading literacy center rotation. She did not receive paraprofessional support in the general education setting. Camilla had a behavior support plan in her IEP. She communicated verbally in one- or two-word phrases and had an approximately 51- to 200-word spontaneous vocabulary. Camilla did not qualify to take her state alternate assessment because she was too young, although her special educator explained she would likely qualify in the next school year. The special educator also reported that Camilla was currently working on learning letter names and sounds. Camilla had literacy-related IEP goals to expressively identify upper and lowercase letters and the main idea of a book she had heard read aloud. She received literacy intervention in the special education classroom using an alternative special education curriculum. The study occurred during literacy center rotations in the general education class. Prior to the study, Camilla attended the general education class for one partner reading rotation. A second rotation time was added to her schedule in which she participated in the shared reading intervention.

Setting and Implementer

One suburban elementary school in a Midwestern state served as the location for this study. The school was the district center for students with autism and communication support needs and had three classes designated for students with autism and communication support needs. Both participants attended this specialized program, which was not at their neighborhood school. All activities and sessions occurred in first- (Camilla) or fourth-grade (Ben) general education classes with an average of 17 students during ongoing literacy centers. The general educator was the only adult in the room other than the implementer, who was the first author of the study. Sessions occurred one-to-one with the implementer and participant. The implementer had 5 years of teaching experience with students with ESN in general education settings. Camilla read at a carpet near peers reading silently, and Ben read at his desk in a pod of four during silent reading.

Materials

Ten nonfiction grade-level books were selected for each participant from the ReadingA-Z.com level that represented the grade level of their general education class to ensure equivalence. The books were randomly selected from all available nonfiction titles at the designated reading level for each student. Student interest or background knowledge was not taken into consideration for book selection or creation of modifications. Camilla's books were level I (Lexile 430-530), and Ben's books were level U (Lexile 820-1030; MetaMetrics, 2004). ReadingA-Z.com books were utilized as they were part of the school literacy curriculum. Books were printed in color on 432 mm by 279 mm paper which left approximately 50.8 mm of blank margin in which modifications could be added. Time to prepare materials for the nonmodified book condition was approximately 20 min per book and 1 hr per book for modified books.

Text Modifications

In the modified condition, the implementer modified books by gluing in pictures and modified text lines into the margins of the typical books. The within-book modifications used in this study were intended to supplement the typical grade-level book rather than replace it. Each modified book included a repetitive text line that summarized the main idea and three or four adapted text sections that highlighted key ideas or vocabulary. In addition to within-book modifications, a prereading concept sheet was created for each book that contained three to four images and each of the adapted text sections that were added within the book.

Students also had access to response options to questions. Ben received a one-page question response sheet with five multiple-choice questions, each with three options. Camilla received five multiple-choice comprehension questions, each with three options, that were embedded directly within the modified book. Examples of modified and nonmodified books for each student are available on Open Science Framework (OSF; https://bit.ly/38eLflP).

Dependent Variable and Response Measurement

The dependent variable was student comprehension of grade-level nonfiction books as measured by independent, accurate responses to comprehension questions after a book was read aloud. Each book had five opportunities to answer comprehension questions during each session. Comprehension questions answered accurately and independently within 5 s of the initial cue were recorded as correct. Responses were recorded incorrect if the student either gave an inaccurate response or failed to respond within 5 s of the end of the question. Ben and Camilla answered questions orally or by pointing to answer options which were only available during the modified book condition. All five comprehension questions were "wh" style (who, what, where, when, why, how) and directly related to content in each nonfiction book.

Experimental Design

A repeated acquisition (RA) single-case research design (Ledford & Gast, 2018) was used to evaluate the impact of modified books as tools to improve the accuracy of student comprehension compared with nonmodified books. The RA design is one of a limited number of single-case designs that assess comparisons between two interventions or conditions for teaching the acquisition of academic skills expected to be acquired quickly (e.g., immediately after reading; Kirby et al., 2021; Ledford & Gast, 2018). Repeated acquisition designs have been used in recent research on literacy interventions (Dennis, 2016; Sullivan et al., 2013) in which multiple sets of equivalent skills are repeatedly measured through pretest and posttest probes (Ledford & Gast, 2018). Repeated acquisition designs are recommended for use when evaluating skill acquisition across targets that are quickly acquired such as information obtained after book reading (Kirby et al., 2021). Repeated acquisition designs also allow for intrasubject replication (Ledford & Gast, 2018). In other words, the quick alternation of interventions in this design provides for multiple comparisons of effect between the two interventions for a single participant. Conducting multiple RA designs in this study allows for intersubject replication across participants. True randomization using a random number generator was used to assign books to conditions (modified or nonmodified shared reading) and identify the order in which each book would be read within each condition (e.g., order of books in modified sessions). Block randomization (two sessions per block) was used to randomly assign sessions to conditions. Recommended quality indicators of the internal and external validity of RA designs were met in this study; internal validity was strengthened through use of randomization (Kirby et al., 2021).

Structured visual analysis was used to identify the presence or absence of a functional relation between book modifications and a change in book comprehension (Ledford & Gast, 2018). Following RA visual analysis guidelines issued by Kirby and colleagues (2021), the change in level (percentage correct responses) between pretest and posttest within each condition was first analyzed (five replications per condition). The consistency of change in comprehension scores between pretest and posttest data was compared within the modified and nonmodified shared reading sessions. Then, comparisons of level changes from pretest to posttest scores across modified and nonmodified shared reading sessions were conducted to determine if the change in level was consistently greater in the modified or nonmodified condition (Kirby et al., 2021). A functional relation between book modifications and a change in book comprehension was established if the increase in the number of comprehension questions answered correctly was consistently higher in one condition over the other (Kirby et al., 2021; Ledford & Gast, 2018).

Study Procedures

Prestudy. Prior to the beginning of the intervention, the first author conducted prestudy reading sessions with each student to identify the appropriate difficulty level of questions for each student and the intensity of modifications. Prestudy reading sessions lasted 20 to 30 min (seven total for Ben and six for Camilla). The first prestudy reading session began by reading a nonmodified grade-level nonfiction book and asking grade-level comprehension questions. If the student did not respond to two questions in a row or responded incorrectly to five questions in a row, increments of more intensified text modification, more simplified questions, and different response mode options were provided until the student could consistently answer most but not all the questions. Motivational and behavioral support strategies were also trialed at this time. Once an appropriate level of adaptation and motivational support was identified, one more book was read with the student that contained all the characteristics that were used in the study to ensure the student responded to at least three out of the five comprehension questions before the study began. A chart of intervention characteristics was made for each student and adhered to for the duration of the study. A decision-making diagram used to guide book modification decisions is found in Figure 1. The characteristics of intervention chart for each student are available on OSF (https://bit.ly/38eLflP); these characteristics are also summarized in Table 1.

Pretest. Pretest procedures were identical across the modified and nonmodified book conditions and occurred one session before the reading of the book represented in the pretest. A pretest session began when the researcher said,

> Hi (student name). We are going to read this book later. I am going to ask you a few questions about this book before we read it. It's ok if you do not know the answers because we haven't read the book yet!

After the introduction, the researcher asked the five comprehension questions assigned to the book. A 5-s response interval was provided after the presentation of each question. If the student responded accurately or inaccurately within the 5 s, the researcher said, "thank you" and moved on to the next question. If the student did not respond after 5 s, the researcher said, "thank you." This procedure was repeated for each of the five questions.

Shared reading with posttest. The procedures for both the nonmodified and modified book conditions were identical except the use of modified or nonmodified books and response options (see section "Materials" for detailed descriptions). All text on every page was read during the reading procedure, including captions, text boxes, and modified text that had

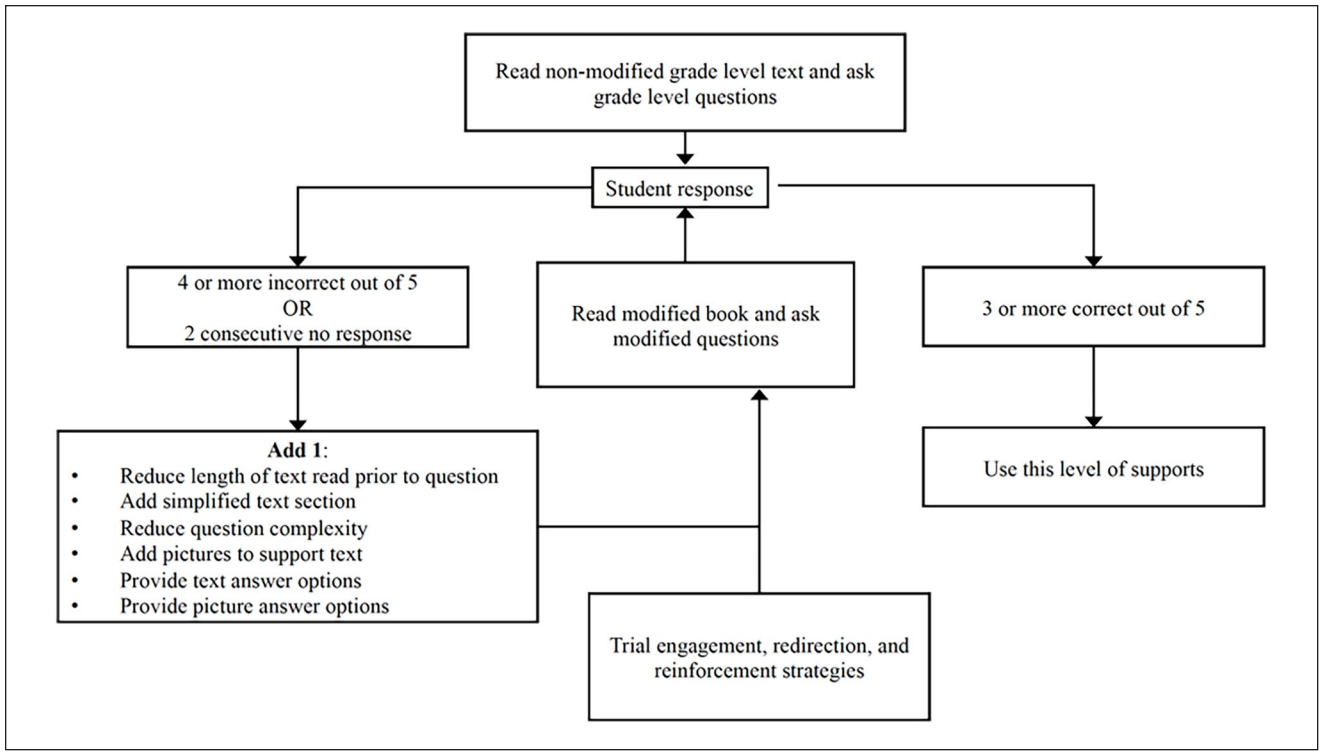


Figure 1. Book modification decision making process.

been added into the book in the modified book condition. In addition, the researcher gained the student's attention by inviting them to interact with each page of the book through pointing, reading, commenting, or turning the page. The reading procedure was paused at five points during each session to ask a posttest question; posttest questions were delivered on the page in which the content needed for accurate answers was provided. The transition from the shared reading procedure to the posttest procedure was cued by stating, "(Student name), I have a question, are you ready?," then stating the question providing a 5-s response interval. If the student responded correctly within 5 s, they received verbal praise. If the student answered incorrectly or did not respond within 5 s, the correct answer was stated. No questions were asked outside of the five preplanned comprehension questions. The transition from the posttest procedure back to the reading procedure was indicated by stating, "Let's keep reading." This process repeated until the last page of the book was read or the last question was answered. Sessions were designed to fit within the naturally occurring 15- to 20-m-center rotations.

Interobserver Agreement and Procedural Fidelity

Secondary observer data were collected during 40% of sessions for all dependent variables across all conditions and participants via video by a trained second-rater for the purpose of estimating interobserver agreement (IOA; Ledford & Gast, 2018). Interobserver agreement was calculated point-by-point using the following formula (agreements/[agreements + disagreements] × 100]). Mean IOA for Ben was 100% across all conditions. Mean IOA for Camilla was 100% for nonmodified pretest, 90% (range, 80%–100%) for modified pretest, and 100% for all posttest conditions. Disagreement occurred on one question in the modified pretest resulting in one session with an IOA of 80% (agreement on four out of the five questions). Both raters reviewed the video of the session and quickly came to consensus on the accurate rating of Camilla's response.

Procedural fidelity (PF) data were collected in 40% of all sessions across all participants and conditions using direct, systematic, observational recording via video with a trial-based behavioral checklist (https://bit.ly/38eLflP) to measure the extent to which conditions were implemented as intended. Mean PF for nonmodified condition pretest sessions was 100% for Ben and 97.6% (range, 95.5%–100%) for Camilla. The error in PF within the nonmodified pretest condition for Camilla occurred when the researcher did not say "thank you" after the student provided an answer in one opportunity. Procedural fidelity for modified condition pretest sessions was 100% for both participants. Procedural fidelity for nonmodified condition posttest sessions was 100% for Ben and 93.9% (range, 87.4%–100%) for Camilla. Procedural fidelity for modified condition posttest sessions was 100% for Ben and 95.1% (range, 90%–98.4%) for Camilla. The most commonly missed steps in both the nonmodified and modified posttest shared reading sessions for Camilla were not

Table 1. Modified Book Characteristics.

Characteristic	Description
	Camilla
Pre-Reading Sheet	Pre-reading vocabulary/concept sheet that contains three images to support main idea of text and all five adapted text lines. Images are color photographs with minimal background distraction (e.g. a plane in the sky rather than on a runway with other vehicles around). This sheet is read aloud prior to reading the book.
Repetitive Text Lines	Repetitive text line with three to four words (no more than 1 new content word) added to every page of book. On first occurrence of line implementer tells student it will repeat, and they will read it. Implementer reads line aloud and asks student to repeat. On subsequent occurrences of the repetitive text line, and after receiving a prompt to read the line, student is given opportunity (at least five seconds) to attempt reading repetitive line independently before receiving assistance to read the line.
Adapted Text	Three or four single sentence adapted text lines added as captions to three or four existing pictures in each modified condition book. Adapted text lines are highlighted in yellow and contain answers to planned comprehension questions within the single sentence. Adapted text lines are always related to main topic of the book/page.
Added Images	No additional images added. Images native to the book used to support comprehension.
Questions	The questions text is embedded in each book and has a yellow background and three single-word or two-word text response options. Questions are embedded in the text to decrease likelihood of losing student attention when introducing a separate sheet of paper. Questions are directly related to the repetitive text line or adapted text section to control for difficulty. Questions and response options are visible to the student while they are read aloud. Response options are pointed to as they are read aloud.
Behavior and Motivation Support	Student appreciates opportunity to re-answer questions she has initially answered incorrectly. If the student answers incorrectly and indicates she would like to try again after the implementer has delivered the controlling prompt, the implementer will read the questions and answer options again to provide a second opportunity (No data is collected on this response). This support is provided to reduce frustration and promote participation. Student receives a fist bump after reading each repetitive text line. Student is given the opportunity to turn each page of the book or interact by pointing to something on each page before reading begins.
	Ben
Pre-Reading Sheet	Pre-reading sheet that contains three vocabulary words that support main idea or concepts covered in the text. Each word is supported by one to three images and two to four sentences that convey the meaning or concept. Images are color photographs or realistic drawings. This sheet is read aloud prior to reading the book. The pre-reading text and images are embedded in the modified book and connected to comprehension questions.
Repetitive Text Lines	Repetitive text line with no more than two short sentences. Lines are written using familiar vocabulary and many decodable words. Lines are written so student can read them independently on second or third occurrence. Repetitive line occurs on every other page.
Adapted Text	Adapted text sections summarized the main idea or a central concept on the pages they were found. Adapted text mirrored text found on the pre-reading sheet.
Added Images	Added imaged were identical those found on the pre-reading sheet. Images were placed directly above adapted text lines and repetitive text lines.
Questions	Student has a one-page question response sheet with five multiple-choice Wh-style questions, each with three, one-to-three-word answer options. Questions are directly related to the repetitive text line or adapted text section to control for difficulty. Questions and response options are visible to the student while they are read aloud. Response options are pointed to as they are read aloud.
Behavior and Motivation Support	Student receives a fist bump after reading each repetitive text line. Student is given the opportunity to turn each page of the book or interact by pointing to something on each page before reading begins.

inviting her to engage with the cover page or table of contents, and not thanking her for attending the session.

Social Validity

Following the completion of the study, the special and general educators for both participants completed a four-question Likert-type style (1 = *strongly disagree*, 6 = *strongly agree*) social validity survey and a semi-structured in-person interview. Survey and interview questions can be obtained on OSF (https://bit.ly/38eLflP). Teachers were presented with all study materials, graphed student response data, and a description of the material creation process prior to their completion of the survey and interview. The goal of the survey and interview was to investigate the extent to which each educator felt the intervention would be feasible in the class and meaningful to student learning. Questions asked teachers to identify the

acceptability, feasibility, effectiveness, and likelihood of implementation of literacy intervention for students with ESN in general education classrooms.

Results

Each participant engaged in 10 shared reading sessions that included a pretest and shared reading with embedded posttest. Pretest for Ben lasted an average of 2 min 12 s (range, 1 min 49 s to 2 min 37 s). Pretest for Camilla lasted an average of 2 min 34 s (1 min 12 s to 11 min 25 s). The first prereading session lasted 11 min and 25 s because Camilla became frustrated that she was asked questions before reading the book and was not told if the answers were correct. This was a new activity structure for her. Camilla was given a break in the middle of the session to provide space to deescalate. She only needed this extra time during the first session. Shared reading and embedded posttest occurred simultaneously. The nonmodified book shared reading sessions for Ben lasted an average of 9 min 57 s (range, 8 min 45 s to 13 min 41 s) and an average of 11 min 3 s (6 min 51 s to 14 min 35 s) for Camilla. The average duration for shared reading and posttest in the modified book reading condition was 16 min 29 s. The modified book shared reading sessions for Ben lasted an average of 17 min 56 s (range, 7 min 12 s to 20 min 43 s) and an average of 13 min 12 s (11 min 11 s to 15 min 40 s) for Camilla. Data are displayed in Figures 2 (Ben) and 3 (Camilla).

Ben

The number of correct responses to comprehension questions across conditions for Ben is displayed in Figure 2. Ben answered zero questions correctly in all pretest sessions in the nonmodified book condition. During the nonmodified book with posttest sessions, Ben's accurate responses remained at zero in three sessions, increased by one in one session, and increased by two in one session. In the modified book condition pretest sessions, Ben answered zero questions correctly in four pretest sessions and one question correctly in one pretest session. During the modified book pretest to posttest sessions, Ben's accurate responses increased by five correct in one session, four correct in three sessions, and three correct in one session. The level and consistency of change in Ben's comprehension scores from pretest to posttest were compared between five paired modified and nonmodified book sessions. Ben answered more comprehension questions correctly in the modified book condition from the pretest to posttest sessions in all five direct comparisons. A functional relation between modified books and correct comprehension question responding was established by Ben's achievement of a larger increase in comprehension score during the modified shared book reading in five out of the five direct comparisons of the modified and nonmodified book conditions.

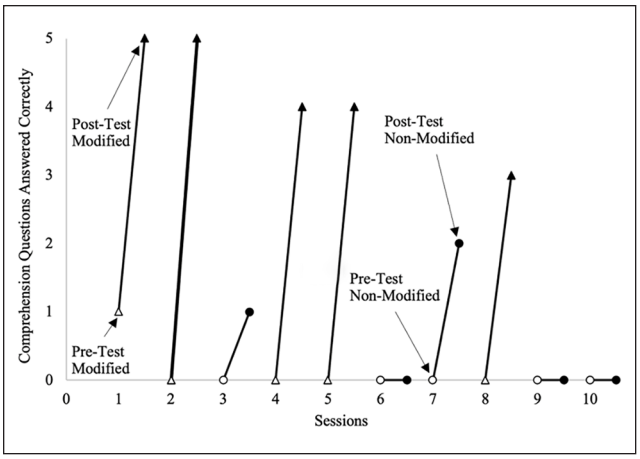


Figure 2. Comprehension results for Ben.

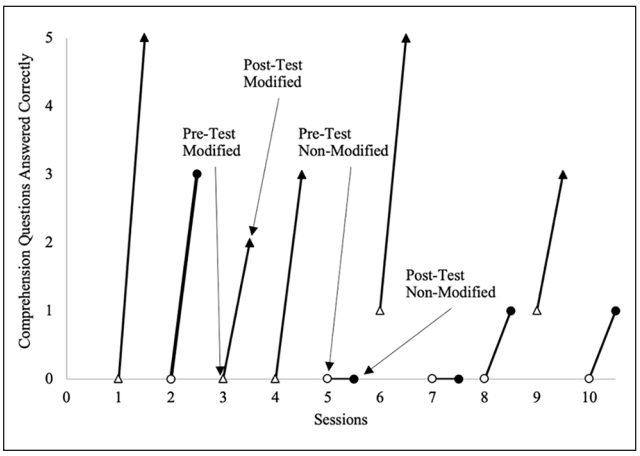


Figure 3. Comprehension results for Camilla.

Camilla

The number of correct responses to comprehension questions across conditions for Camilla is displayed in Figure 3. Camilla answered zero questions correct in all pretest sessions in the nonmodified book condition. During the nonmodified book posttest sessions, Camilla's accurate responses remained at zero in two sessions, increased by one in two sessions, and increased by three in one session. In the modified book pretest sessions, Camilla answered zero questions correct in three pretest sessions and one question correct in two sessions. During the modified book posttest sessions, Camilla's accurate responses increased by five questions in one session, four in one session, three in one session, and two in one session. The level and consistency of change in Camilla's comprehension scores from pretest to posttest were compared between five paired modified and nonmodified book sessions. Camilla answered more comprehension questions correctly in the modified book condition from the pretest to the posttest in all five

direct comparisons. Although there was minimal overlap in the number of correct responses from Camilla between conditions, a functional relation between modified books and correct response was established by Camilla's achievement of a larger increase in comprehension score during modified shared book reading in five out of the five direct comparisons of the modified and nonmodified book conditions.

Social Validity

The two general and two special educators who supported Ben and Camilla completed a social validity survey and interview. Their responses are summarized in the following section, and a table of all teacher responses to the social validity survey can be obtained on OSF (https://bit.ly/38eLflP). All teachers agreed that students with ESN can learn academic content in general education settings (M = 5.5; range, 4–6) and that the intervention supported access to general education curriculum (M = 5.5; range, 5–6) and progress in individualized learning goals (M = 5.0; range, 4–6). All educators strongly agreed that participant support needs were extensive enough to warrant the use of modified grade-level books to facilitate access to content (score 6.0) and they would suggest the intervention to others (M = 5.25; range, 4–6). During the in-person interview, educators cited academic, social, and emotional benefits of shared reading in the general education setting. Camilla's special educator explained, "She wasn't just a body in the classroom. There was somebody there with her that was doing academic kinds of things with her." The special educator saw this as more academically beneficial than the 15 min per day that Camilla attended the general education class for partner reading. Camilla's general educator agreed stating, "I think when they partner read she would just sit and the other kids would be reading . . . but with this one [shared reading], she was actually participating and answering questions." Ben's special educator felt "it gave [Ben] confidence being able to be in class and reading a book." He explained that Ben typically completes worksheets sent from the special education classroom during silent reading time in the general education classroom, and the special educator could tell Ben was excited to read a book while other students were also reading. Ben's general educator added that it was good for her students to see Ben reads books because now that the study is over, "two days [Ben] is now working with [fourth-grade student]. . .and they read together, and he seems to really enjoy that too."

All educators reported they could see themselves or a paraprofessional implementing the intervention (score 5.0) but expressed specific barriers. Educators shared that the biggest barrier to using shared reading in general education settings was the out of school time and cost to create materials, as well as the instructional time needed to implement shared reading one-on-one. Ben and Camilla's special educators both explained they could see themselves creating the individualized modified books, but it would not be feasible with their current workloads. Ben's special educator added that "the intimidating part is doing it week by week. I would need to sit down and just knock out, say, 50 stories over the summer, and that way, I can just start pulling things for kids at different grade levels." Both general educators felt material creation would be time-consuming, but expressed they felt that time commitment would be worthwhile to support student access. Ben's general educator stated, "It is like anything else, if you have any other student with low needs or high needs in the classroom, you're still making supplemental work for them too. So it's no different for [Ben]." Both general educators stated printing the books as a barrier because they have to pay for color printing. Special educators expressed they could print materials due to fewer restrictions on color printing.

When asked if educators felt they would use shared reading in the future, all educators responded they felt it would be difficult for them to deliver an intervention to just one student in the general education classroom. General educators cited the need to support other students in their class, and both special educators explained that they are not able to leave their special education classrooms. All interviewees felt the intervention would be most feasible for a paraprofessional to deliver in the general education classroom. Camilla's special educator explained,

> Ideally, I would be able to train a para to do it . . . I could see myself doing that with her if staffing were different in these programs. If I leave and I leave the classroom so that there's going to be five children with only two adults . . . and the kids that are left in here are really challenging. That's the barrier, right there, staffing.

Ben's general educator mentioned that the intervention would be more feasible for her to implement if it were included as part of group instruction. She said, "you (the first author) did one-on-one, and that time in the classroom, that's very valuable to me, that's why I think a group setting is so much better because you're at least attending to several students instead of just one." In summary, educators felt the intervention should be implemented by paraprofessionals or adapted to occur during group instruction delivered by the general educator.

Discussion

Findings from this study indicate two students with ESN were able to respond to comprehension questions more accurately during shared book reading when grade-level nonfiction books were provided with text and question structure modifications in general education classrooms. These findings add to the growing base of support for

shared reading and modified books to support comprehension of students with ESN in general education settings.

The text modifications (a combination of a prereading concept review sheet, simplified text sections and repetitive text lines embedded in grade-level nonfiction books, and questions with individualized response options) used during shared reading were effective in supporting higher nonfiction book comprehension for Ben and Camilla. The components of modified books and the decision-making diagram used in this study may be useful for making a variety of curricular areas more accessible through content area reading activities. Modification of nonfiction books with individualized response options within shared reading instruction has been successfully provided students with ESN access to science (Wood et al., 2019) and social studies (Wood et al., 2015) instruction in general education classes.

A contribution of this study is the explicit identification of a modification decision-making process (see Figure 1) and the placement of text modifications within the margins of grade-level books (see examples on OSF; https://bit.ly/38eLflP). Although research exists on the benefits of book modifications to support content knowledge and skill acquisition, there is little guidance on how educators can systematically identify the level of modification that will be most effective (Kurth & Keegan, 2014; Roberts & Leko, 2013). Nonfiction book modification for students with ESN represents a valuable strategy for educators to meet federal requirements of providing access to general education curriculum (U.S. Department of Education, 2004) while simultaneously supporting literacy and content area knowledge development for students.

Educators indicated that the shared reading in the general education settings seemed meaningful and beneficial academically, socially, and emotionally for the participants with ESN. These sentiments are echoed by research that has associated the education of students with ESN in general education settings with (a) higher learning expectations; (b) increased engagement, participation, social interactions, and access to general education curriculum; and (c) improved academic, communication, and social skills (Carter, 2017; Kleinert et al., 2015; Kurth & Mastergeorge, 2012). In fact, students with ESN are 10 times more likely to access literacy instruction in general education settings than in special education settings (Ruppar et al., 2018).

Although educators reported benefits of shared reading, all expressed it would be difficult to implement a 20-min intervention with only one student within the general education classroom. General educators suggested the integration of the intervention within group instruction to improve the feasibility of the intervention. Special educators shared they are not able to provide instruction in the general education setting within the current staffing structures at their campuses. General and special educators shared an additional barrier to the implementation of this intervention in the 1-hr preparation time for each modified book lesson and the cost associated with printing materials in color. The educators' positive impression of the intervention indicates that with a restructuring of existing resources and roles within their school, literacy instruction in general education settings for students with ESN, such as one-on-one shared reading, may be more feasible (McCart et al., 2014). This restructuring should facilitate special educators' ability to spend time delivering instruction and support to students in general education settings. In addition, educators should be provided time and compensation for collaborative planning to prepare accessible and individualized supports for their students.

Limitations and Future Directions for Research

There are several limitations to consider when interpreting the results of this study. This intervention was not implemented by an authentic intervention agent such as a paraprofessional, peer, general, or special educator. There is a need for research implemented by authentic intervention agents in typical student groupings to advance knowledge about the possible efficacy and feasibility of shared reading with modified books in general education settings. Future research on shared reading should focus on (a) the supports necessary for authentic intervention agents to implement shared reading for students with ESN in general education, (b) student learning outcomes of shared reading in general education settings, and (c) how shared reading can be integrated into small and whole group instruction.

Although there is an existing base of support for shared reading, most studies have been conducted in part or whole within separate special education settings rather than general education settings (Toews et al., 2021; Hudson & Test, 2011). This study represents only the third study of the impact of shared reading on literacy skills of students with ESN that has been conducted entirely in the general education setting. It adds to the growing base of knowledge on the efficacy of embedding shared reading within general education settings to support students with ESN to read nonfiction books. However, more research is needed to understand the most effective and efficient methods for delivering shared reading in inclusive settings.

Furthermore, despite this RA design meeting internal and external design standards, additional research replicating the evaluation of text modifications also should be conducted. Future research should also be designed to include a measure of maintenance or generalization of comprehension skills to evaluate the impact on student general skills and knowledge. This measure was lacking in this study. This study introduced modifications to the books and comprehension questions in the posttest phase which may affect the internal validity of the study. Future research should investigate the impact of the introduction of these

two modification types separately to better understand their utility. In addition, each student required individualized modifications; thus generalization of these findings for every student with ESN is impractical. However, additional evaluations of the process for selecting and implementing text modifications can inform guidance for selecting individualized literacy supports for students with ESN in inclusive settings. Finally, this study measured comprehension through only the use of multiple-choice comprehension questions. Although this is a common formative classroom assessment structure, future research should investigate the same methods with alternative means of measuring comprehension such as question generation, open-ended questioning, or journaling.

Future research designs should include investigation of how shared reading interventions can be integrated into small and whole group instruction. Both Ben and Camilla's general education teachers expressed that they would not be able to spend 20 min multiple times per week teaching a single student. Ben's teacher suggested that she would be more interested in delivering the intervention if it was in a group format using a book that was part of the class curriculum. It is important that future research investigate the use of shared reading using authentic grade-level books in small and whole group instruction to support the individualized learning goals of students with ESN.

Research on systematic methods for the identification of the type and intensity of book modification students need to access and make progress toward their literacy goals is needed. The prestudy reading sessions were integral in the identification of effective student literacy supports. For example, in this study, modified text sections for Ben were two or three sentences, and those for Camilla were only one. A systematic approach to trialing different lengths and complexities of book modification was used until the supports that would appropriately challenge students were identified. Increased research in this area could support practicing educators in the creation of accessible literacy materials. In addition, educator preparation faculty at universities could use research on modification intensity identification to integrate new guidance into their methods courses.

Implications for Practice

Shared reading using modified grade-level nonfiction books may decrease educator need for alternative special education literacy curricula that lack the rich content, scope and sequence, and exposure to deep content discussion students gain from authentic books presented in general education settings (Taub et al., 2017). The results of this study indicate shared reading using modified books, prereading concept review sheets, and preplanned comprehension questions with individualized response options was a valuable strategy for supporting access to and progress on individualized student comprehension goals. The modifications used to support comprehension in this study were embedded directly within typical grade-level nonfiction books. Because the simplified text and repetitive lines are embedded and the whole grade-level book is read aloud, students access the rich content of the grade-level book with supplemental supports to increase comprehension. These findings promote use of modified grade-level books rather than special education curricula to increase access to the general education standards.

Declaration of Conflicting Interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

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